UNIVERSITY OF KELANIYA / SENG 34213

FOREVER HOTEL MANAGEMENT SYSTEM

Test Coverage Report

Six-Subsystem Coverage and Automated Test Execution

Reporting date: 25 October 2026. Evidence is identified by subsystem branch, commit, and run date; comparisons across different revisions are not a single hotel-wide release test.

Contents

1. Consolidated coverage overview

2. Hotel Website (HW)

3. Manager Dashboard (MAD)

4. Front Desk System (FDS)

5. Food Ordering and Service Requests (FOSS)

6. Kitchen Management System (KMS)

7. Worker Management System (WKMS)

8. Evidence interpretation and remaining actions

---

1. Consolidated coverage overview

The table records instrumented source coverage for separately tested components. Values are reported in percent; a dash indicates no measured value in the provided CI evidence. No average across subsystems is calculated because coverage configurations, branches, and included files differ.

Subsystem / measured scope	Statements	Branches	Functions	Lines
HW - Backend Jest source scope	91.03	85.17	95.93	91.82
HW - Frontend selected TypeScript files	Not reported	99.45	100.00	99.48
MAD - Backend Jest/V8	97.08	90.69	92.24	97.08
MAD - Frontend Vitest/V8	94.27	91.25	93.34	96.02
FDS - Backend Jest source scope	96.89	87.19	96.71	96.66
FDS - Frontend Jest source scope	89.77	80.91	92.36	89.67
FOSS - Backend historical develop CI	59.01	59.37	50.00	60.37
FOSS - Backend separate feature CI	69.56	72.22	54.16	70.73
FOSS - Frontend	Not measured	Not measured	Not measured	Not measured
KMS - Backend unit/E2E	Not measured	Not measured	Not measured	Not measured
KMS - Frontend Jest	Not measured	Not measured	Not measured	Not measured
WKMS - Backend Jest source scope	93.28	88.19	89.30	94.80
WKMS - Frontend Jest source scope	93.81	85.55	94.21	94.90

Test execution and traceability

Subsystem	Reported passing automated tests	Selected register cases	Core qualifications
HW	270	33	Frontend coverage reuses 21 selected tests
MAD	1,070 core + 6 Node checks	58	Current run audit gate failed
FDS	858	43	Historical develop CI
FOSS	18 develop / 34 feature (not additive)	42	Different tested branches
KMS	76	82	No measured coverage
WKMS	445	37	33 backend tests skipped

Selected test-register cases are documented examples and acceptance gaps, not the complete raw automated test inventories. Counts refer to each repository snapshot; alternate FOSS runs and the HW coverage rerun are deliberately not added together.

2. Hotel Website (HW)

Repository: forever-hotel/hotel-website | Branch: pre-develop | Tested revision: 606acf0674a18621639bce572ce3577af933cd63 | CI date: 22 October 2026

Evidence: https://github.com/forever-hotel/hotel-website/actions/runs/39726186380, https://github.com/forever-hotel/hotel-website/actions/runs/39726186472, https://github.com/forever-hotel/hotel-website/actions/runs/39726186469

Executed test groups

Test group	Suites	Passed	Failed	Skipped
Backend unit	32	212	0	0
PostgreSQL integration	8	36	0	0
Frontend Node tests	Not specified	22	0	0

Measured coverage

Measured scope	Statements	Branches	Functions	Lines
Backend Jest source scope	91.03	85.17	95.93	91.82
Frontend selected TypeScript files	Not reported	99.45	100.00	99.48

Scope and interpretation

• The four backend Jest thresholds are 80%. The frontend uses Node test instrumentation for selected .ts files; no corresponding global frontend minimum was identified.

• The separate frontend coverage command reran 21 of the 22 frontend tests; those 21 are not added again to the 270 distinct passing tests.

• The source snapshot predates the complete public booking/payment workflow. Coverage is not functional acceptance.

CI and reproduction evidence

https://github.com/forever-hotel/hotel-website/actions/runs/39726186380
https://github.com/forever-hotel/hotel-website/actions/runs/39726186472
https://github.com/forever-hotel/hotel-website/actions/runs/39726186469

Reproduction / configured coverage command: backend: npm run test:cov -- --runInBand; backend integration: npm run test:integration; frontend: npm run test:coverage

3. Manager Dashboard (MAD)

Repository: forever-hotel/manager-dashboard | Branch: development | Tested revision: bba65a0762286d4b0ac902865d7a2dd1e683e9c7 | CI date: 24 October 2026

Evidence: https://github.com/forever-hotel/manager-dashboard/actions/runs/40137091073, https://github.com/forever-hotel/manager-dashboard/actions/runs/39847903811

Executed test groups

Test group	Suites	Passed	Failed	Skipped
Backend unit	38	532	0	0
Backend integration/E2E	16	207	0	0
Frontend Vitest	34	331	0	0
Separate frontend Node precheck	Not specified	6	0	0

Measured coverage

Measured scope	Statements	Branches	Functions	Lines
Backend Jest/V8	97.08	90.69	92.24	97.08
Frontend Vitest/V8	94.27	91.25	93.34	96.02

Scope and interpretation

• The current CI run passed its test and coverage stages but failed overall in backend dependency auditing; an earlier workflow passed completely.

• Global line and branch thresholds are 80%; selected critical modules use 90% branch thresholds and changed-code checking.

• The current run records 1,070 core passed tests. Six separate frontend Node prechecks are additional, not part of core suite count.

CI and reproduction evidence

https://github.com/forever-hotel/manager-dashboard/actions/runs/40137091073
https://github.com/forever-hotel/manager-dashboard/actions/runs/39847903811
Reproduction / configured coverage command: Backend Jest and frontend Vitest coverage plus changed-code checks in .github/workflows/ci.yml

4. Front Desk System (FDS)

Repository: forever-hotel/front-desk | Branch: develop | Tested revision: 503465b7eb2c29b56cc2ec0d329d91d6b40853d4 | CI date: 24 October 2026

Evidence: https://github.com/forever-hotel/front-desk/actions/runs/40057114169, https://github.com/forever-hotel/front-desk/actions/runs/40057114175

Executed test groups

Test group	Suites	Passed	Failed	Skipped
Backend Jest unit	63	479	0	0
PostgreSQL integration	9	53	0	0
Backend HTTP/Socket.IO E2E	10	140	0	0
Frontend Jest	40	186	0	0

Measured coverage

Measured scope	Statements	Branches	Functions	Lines
Backend Jest source scope	96.89	87.19	96.71	96.66
Frontend Jest source scope	89.77	80.91	92.36	89.67

Scope and interpretation

• Coverage relates to the 24 October develop revision, not a rerun on feature/84-fds-final-qa-evidence.

• Backend unit coverage excludes various bootstrap, migration and module files; frontend instrumentation is limited to configured source paths.

• PostgreSQL integration and HTTP/E2E counts are distinct test groups and are not part of the backend unit coverage numerator.

CI and reproduction evidence

https://github.com/forever-hotel/front-desk/actions/runs/40057114169
https://github.com/forever-hotel/front-desk/actions/runs/40057114175
Reproduction / configured coverage command: backend: npm run test:cov -- --runInBand; test:integration; test:e2e. frontend: npm run test:cov

5. Food Ordering and Service Requests (FOSS)

Repository: forever-hotel/food-ordering | Branch: develop / feature/34-booking-room-verification | Tested revision: 17af08411e / 343b7d0018 | CI date: 13 October 2026

Evidence: https://github.com/forever-hotel/food-ordering/actions/runs/38381405941, https://github.com/forever-hotel/food-ordering/actions/runs/38419260802

Executed test groups

Test group	Suites	Passed	Failed	Skipped
Historical develop unit	5	9	0	0
Historical develop E2E	2	9	0	0
Separate feature unit	7	25	0	0
Separate feature E2E	2	9	0	0
Frontend automated	0	Not run	-	-

Measured coverage

Measured scope	Statements	Branches	Functions	Lines
Backend historical develop CI	59.01	59.37	50.00	60.37
Backend separate feature CI	69.56	72.22	54.16	70.73
Frontend	Not measured	Not measured	Not measured	Not measured

Scope and interpretation

• The historical develop and separate feature-branch runs overlap; do not sum their 18 and 34 passing backend tests.

• Feature-branch coverage (70.73% lines; 72.22% branches) remains below the course 80% target; no frontend coverage was reported.

• A later recovery develop revision was noted in the source package without a matching verified CI run; neither tested snapshot is a result for that later revision.

CI and reproduction evidence

https://github.com/forever-hotel/food-ordering/actions/runs/38381405941
https://github.com/forever-hotel/food-ordering/actions/runs/38419260802
Reproduction / configured coverage command: Backend Jest test:cov; frontend coverage not implemented in the documented snapshots

6. Kitchen Management System (KMS)

Repository: forever-hotel/kitchen-management | Branch: develop | Tested revision: 89087002ac2df8f07b6140c00111979c75bc2799 | CI date: 10 October 2026

Evidence: https://github.com/forever-hotel/kitchen-management/actions/runs/38038213541

Executed test groups

Test group	Suites	Passed	Failed	Skipped
Backend unit	23	65	0	0
Backend E2E	2	5	0	0
Frontend Jest	3	6	0	0

Measured coverage

Measured scope	Statements	Branches	Functions	Lines
Backend unit/E2E	Not measured	Not measured	Not measured	Not measured
Frontend Jest	Not measured	Not measured	Not measured	Not measured

Scope and interpretation

• The successful workflow runs tests without --coverage and does not publish coverage percentages or demonstrate an enforced 80% coverage gate.

• E2E primarily covers scaffold and database/health checks; full kitchen/KOT workflows are not demonstrated by the measured test run.

• The baseline is the evaluated develop revision; separate feature-run results are not substituted.

CI and reproduction evidence

https://github.com/forever-hotel/kitchen-management/actions/runs/38038213541
Reproduction / configured coverage command: backend/frontend: npm run test:cov -- --runInBand (not run in the cited CI evidence)

7. Worker Management System (WKMS)

Repository: forever-hotel/worker-management | Branch: develop | Tested revision: 0f91f1c9d6390c80c9f12f82b834fee9bf374104 | CI date: 25 October 2026

Evidence: https://github.com/forever-hotel/worker-management/actions/runs/40174879698

Executed test groups

Test group	Suites	Passed	Failed	Skipped
Backend Jest (includes skipped)	36 passed; 7 skipped	322	0	33
Frontend Jest	20	123	0	0

Measured coverage

Measured scope	Statements	Branches	Functions	Lines
Backend Jest source scope	93.28	88.19	89.30	94.80
Frontend Jest source scope	93.81	85.55	94.21	94.90

Scope and interpretation

• Seven backend suites and 33 backend tests are skipped, not passed.

• The 80% global line and branch thresholds are configured for both backend and frontend. Statement and function coverage is measured but not an independent gate.

• Authenticated Socket.IO server/client code exists at the evaluated snapshot; cross-service delivery and public staging acceptance are distinct unverified evidence.

CI and reproduction evidence

https://github.com/forever-hotel/worker-management/actions/runs/40174879698
Reproduction / configured coverage command: backend: npm run test:cov -- --runInBand; frontend: npm run test:coverage

8. Evidence interpretation and remaining actions

• Archive dated HTML, LCOV, test-log and CI artifacts before temporary artifacts expire, ensuring the artifact SHA matches the cited branch.

• Run the complete matrix on the approved release candidate. Confirm all subsystems use the same integrated version and record failed/skipped results.

• Complete missing KMS and FOSS frontend coverage, reconcile below-target FOSS coverage, and retain documented exclusions and configured coverage gates.

• Report 80% new-code and critical-branch coverage only where the corresponding changed-code or per-module gate has actually been checked.

• Keep the result of the current Manager Dashboard audit gate distinct from its passing test and coverage stages.

The reported functional correctness of isolated unit, integration, or mock suites must not be treated as completed end-to-end acceptance. User acceptance, production deployment, and complete security testing require separate dated evidence.